

Markscheme

Mathematics: analysis and approaches
Practice question bank

155. (a)

The equation $x^2 + bx + c = 0$, where b and c are real, has a root $x = 2 + 3i$.

Find the value of b and the value of c .

the other root is $2 - 3i$; sum of roots $= 4 = -b$ (M1)

$b = -4$ A1

product of roots $= c = |2 + 3i|^2 = 4 + 9$ (M1)

$c = 13$ A1

[4 marks]